

Supplementary Material

From roughness uncertainty to model-structure effects in 2D flood inundation models: a Monte Carlo comparison of SFINCS and HEC-RAS

Robustness of main conclusions to inter-class Manning correlation. The following figures and tables document the Gaussian copula importance-sampling procedure and its results in full, supporting the correlated-roughness robustness analysis reported in the main manuscript.

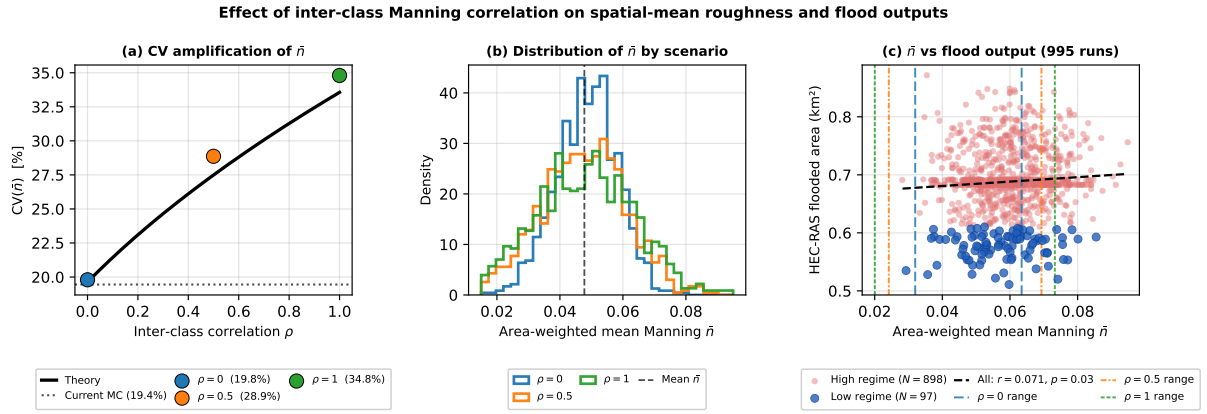


Figure S1: Effect of inter-class Manning correlation on spatial-mean roughness and flood outputs ($N = 1000$ Gaussian copula samples per scenario). (a) Theoretical CV amplification of the area-weighted mean Manning $\bar{n}$ with equi-correlation ρ (solid curve); markers show empirical values; dotted line marks the reference independent-MC ensemble. (b) Step histograms of $\bar{n}$: the distribution widens from CV = 19.8% ($\rho = 0$) to 34.8% ($\rho = 1$) with no shift in the mean ($\bar{n} \approx 0.048 \text{ m}^{-1/3}\text{s}$). (c) $\bar{n}$ against HEC-RAS flooded area for 995 valid simulations coloured by hydraulic regime (upper-left legend: 5th–95th percentile range of $\bar{n}$ per scenario; lower-right legend: regime counts and regression); $r = 0.071$ ($p = 0.03$, $r^2 < 0.01$) shows only a weak direct association between inter-class Manning correlation and domain-scale flooded area; the regime fractions under correlated sampling are quantified in Supplementary Figure S3.

**Manning inter-class correlation structure for three sampling scenarios
(1 000 combinations each; three land use class pairs)**

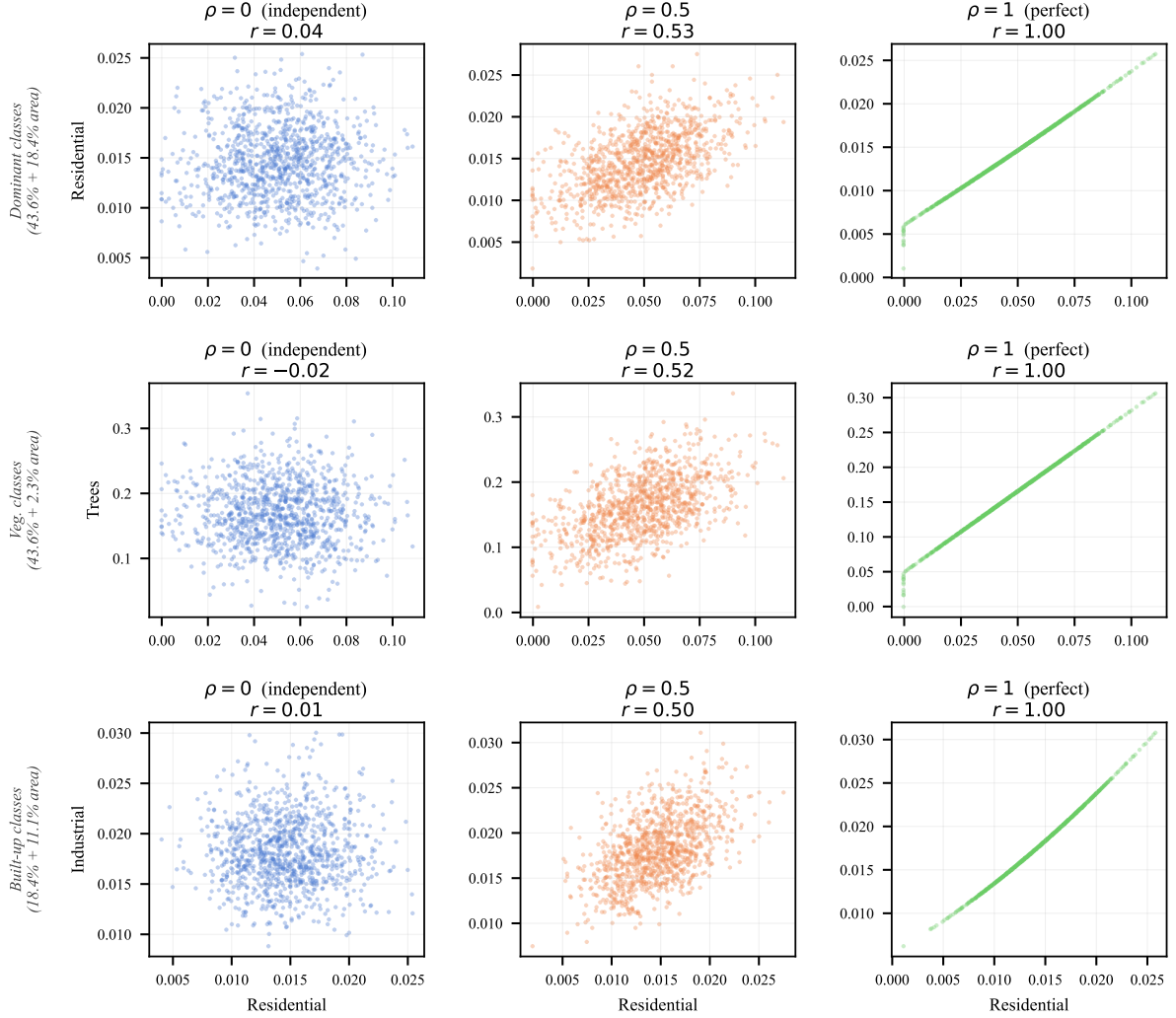


Figure S2: Pairwise scatter plots of Manning n values for three representative land use class pairs and three correlation scenarios ($N = 1000$ Gaussian copula samples each). Rows correspond to class pairs (dominant: sparse vegetation + residential; vegetation: sparse vegetation + trees; built-up: residential + industrial); columns correspond to equi-correlation values $\rho \in \{0, 0.5, 1.0\}$. Empirical Pearson r confirms the copula reproduces the prescribed dependence structure.

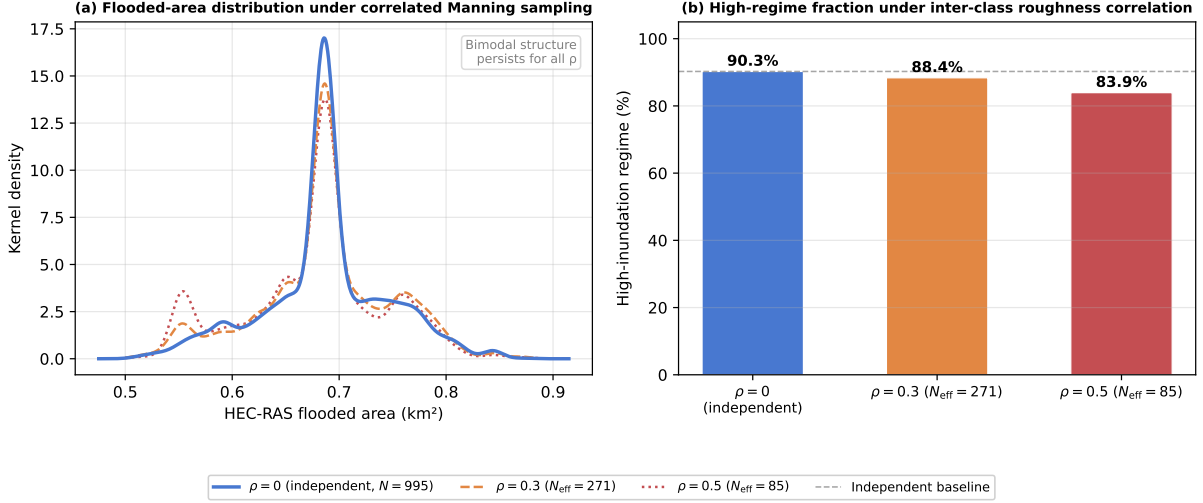


Figure S3: Importance-sampling robustness check for inter-class Manning correlation. **(a)** Weighted kernel density estimates of HEC-RAS flooded area under independent sampling ($\rho = 0$, $N = 995$) and two correlated scenarios ($\rho = 0.3$, effective $N = 271$; $\rho = 0.5$, effective $N = 85$). Weights are the ratio of the equicorrelation Gaussian copula density to the independent sampling density, computed from empirical probability integral transforms of the 9-class Manning vector. The bimodal structure of the HEC-RAS response is preserved for all values of ρ shown. **(b)** High-inundation-regime fraction (weighted percentage) as a function of equicorrelation ρ . The fraction decreases from 90.3% (independent) to 88.4% ($\rho = 0.3$) and 83.9% ($\rho = 0.5$), confirming that the bimodal response is robust to plausible levels of inter-class roughness correlation.

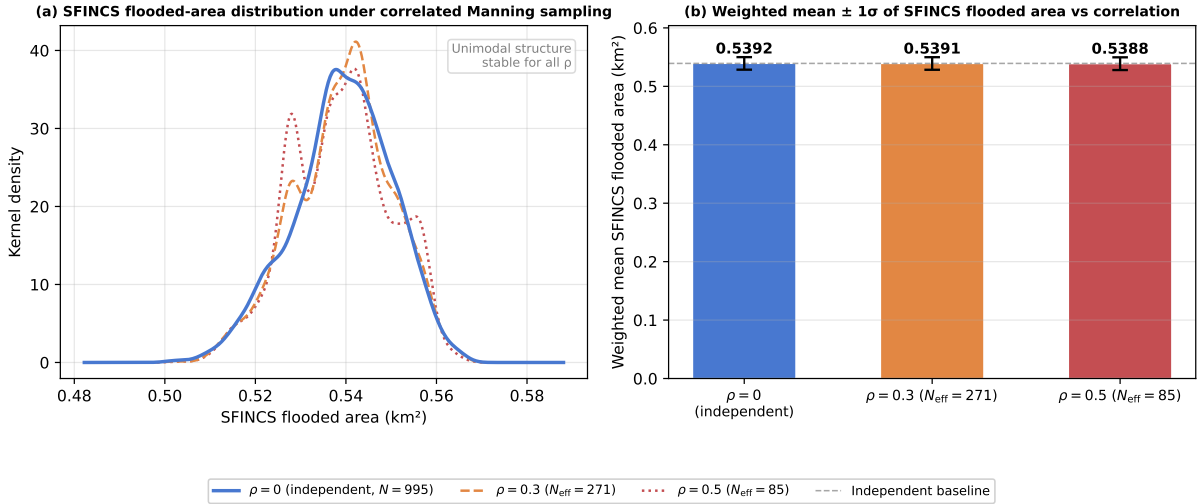


Figure S4: Importance-sampling robustness check for SFINCS under inter-class Manning correlation (counterpart to Supplementary Figure S3 for HEC-RAS). **(a)** Weighted kernel density estimates of SFINCS flooded area under independent sampling ($\rho = 0$, $N = 995$) and two correlated scenarios ($\rho = 0.3$, effective $N = 271$; $\rho = 0.5$, effective $N = 85$). The unimodal distribution is virtually unchanged across all values of ρ . **(b)** Weighted mean $\pm 1\sigma$ of SFINCS flooded area as a function of equicorrelation ρ . The mean shifts by less than 0.1% between $\rho = 0$ (0.5392 km²) and $\rho = 0.5$ (0.5388 km²), confirming that the unimodal SFINCS response is robust to plausible levels of inter-class roughness correlation.

Table S1: Area-weighted spatial mean Manning $\bar{n}$ statistics under independent and Gaussian-copula correlated sampling ($N = 1\,000$ draws per scenario; domain area 1 086 ha). P_5 and P_{95} denote the 5th and 95th percentiles.

Scenario	ρ	Mean $\bar{n}$	CV(%)	P_5	P_{95}	P_{95}/P_5
Independent (reference)	0	0.0484	19.4	0.033	0.064	1.94
Copula $\rho = 0$	0	0.0486	19.8	0.032	0.063	1.98
Copula $\rho = 0.5$	0.5	0.0474	28.9	0.024	0.069	2.87
Copula $\rho = 1.0$	1.0	0.0474	34.8	0.020	0.073	3.66

Table S2: **Simulations removed prior to analysis** ($N = 5$, 0.5% of 1000). Anomalies were detected using a robust z -score with threshold $k = 5$. Model: S = SFINCS; H = HEC-RAS. Four of the five anomalies were flagged in HEC-RAS; the SFINCS outputs for those four simulations lie within the normal ensemble range. Simulation 633 is the exception: the flagged anomaly originates in SFINCS (flooded area = 0.467 km²), while the paired HEC-RAS output falls within the normal ensemble range. Ensemble medians for reference: HEC-RAS area ≈ 0.668 km², depth ≈ 1.22 m; SFINCS area ≈ 0.537 km², depth ≈ 1.04 m.

Sim.	Model	Flagged metric	Flagged value	$ z _{\text{robust}}$	HEC-RAS output		SFINCS output	
					Area (km ²)	Depth (m)	Area (km ²)	Depth (m)
29	H	Mean depth	1.930 m	6.70	2.187	1.930	0.498	0.949
295	H	Mean depth	1.772 m	5.21	1.664	1.772	0.499	0.948
633	S	Flooded area	0.467 km ²	6.84	0.865	1.545	0.467	1.005
724	H	Mean depth	1.805 m	5.52	1.118	1.805	0.512	0.994
755	H	Mean depth	1.834 m	5.79	1.865	1.834	0.503	0.962